

Unit 9, Lesson 10: Similar Triangles and Dilations in the Real World

Benchmark

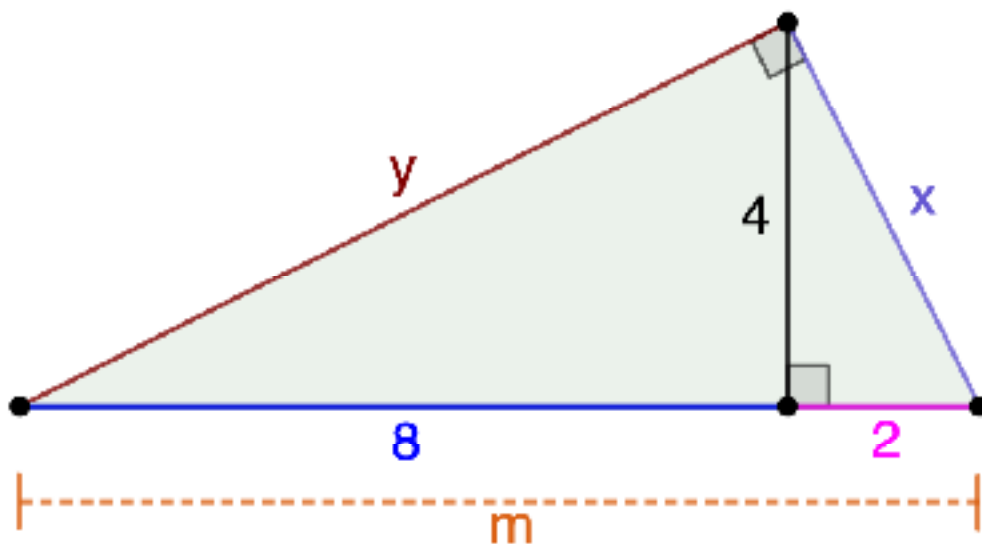
MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

Learning Target

☐ I can solve real-world problems that involve similar triangles.

9.10.1 Warm-Up: Relationships in Triangles

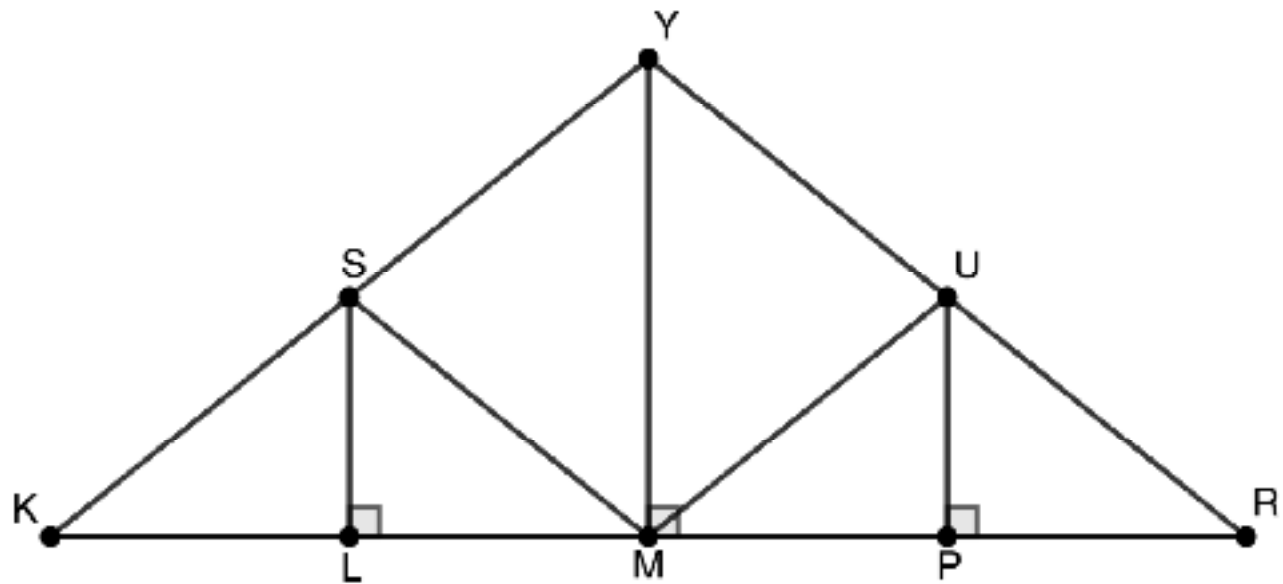
1. A figure is shown. Side lengths and distances, in units, are labeled using variables and known quantities.



- a. How many right triangles do you see in the figure?
- b. Write at least two equations that relate some of the side lengths or distances marked in the figure.

9.10.2 Guided Practice: Dilations and Similar Triangles in the Real World

The Waddell "A" Truss Bridge was built in 1898 and connects Parkville, MO to Trimble, MO. Built using steel rods, the bridge was originally used by the railway but is currently a pedestrian bridge.



1. A scale drawing of the bridge is shown.

The total span of the bridge is 30.5 meters (m), and the height at its highest point is 12.2 m. The span of a bridge is the length of the bridge.

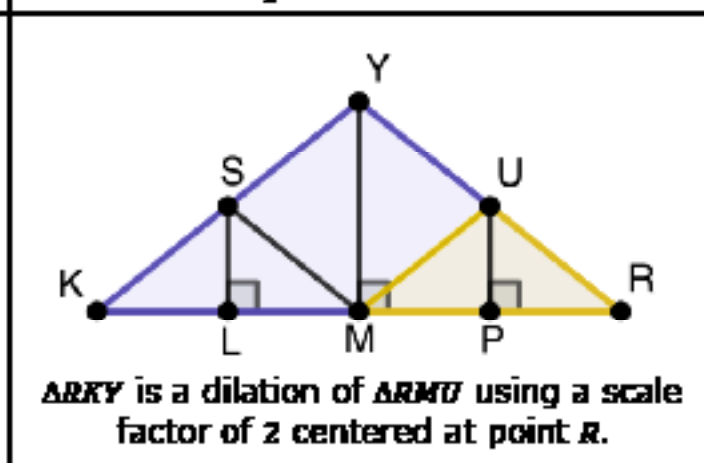
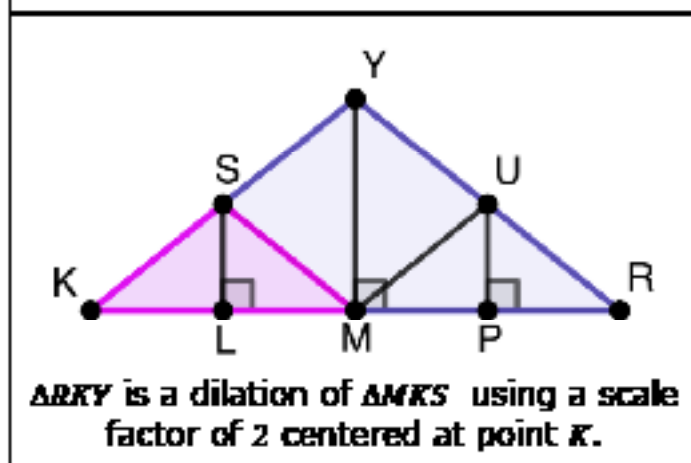
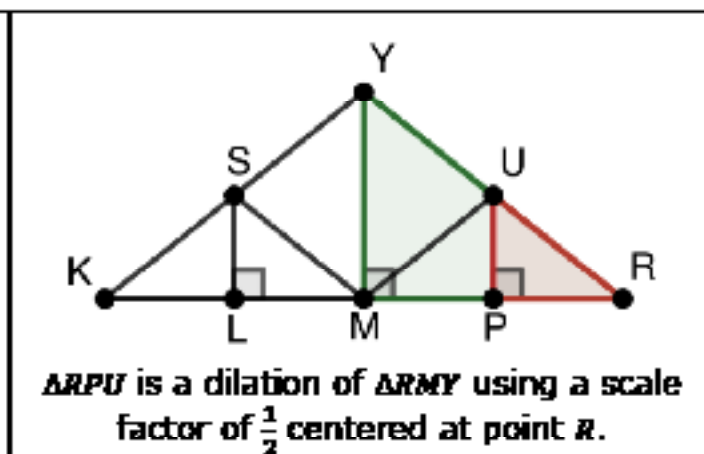
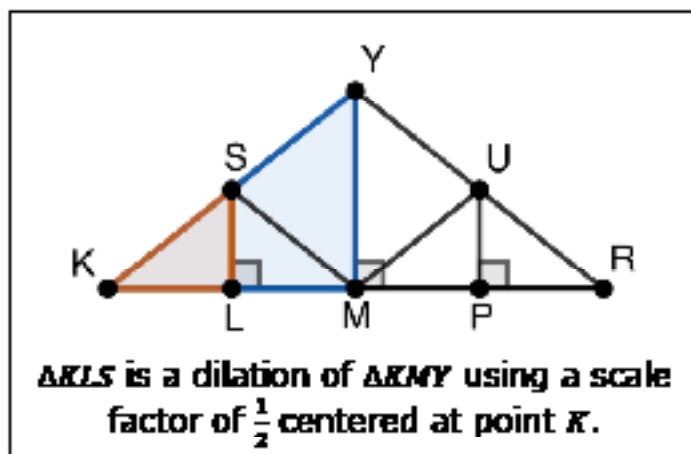
- a. Label the given measurements on the figure.

- b. Complete each statement using the given information.

The span of the bridge is represented by line segment _____.

The height of the bridge is represented by line segment _____.

The images and statements below give additional information about the bridge design.



- c. Use the Pythagorean Theorem, properties of similar triangles, and other reasoning to find the lengths of the line segments. Round to the nearest thousandth of a meter.

Line Segment	Work/Reasoning	Length (m)
$\overline{KR}$	Given	30.5
$\overline{YM}$	Given	12.2
$\overline{SL}$		
$\overline{UP}$		
$\overline{YK}$		
$\overline{YR}$		
$\overline{SM}$		
$\overline{UM}$		

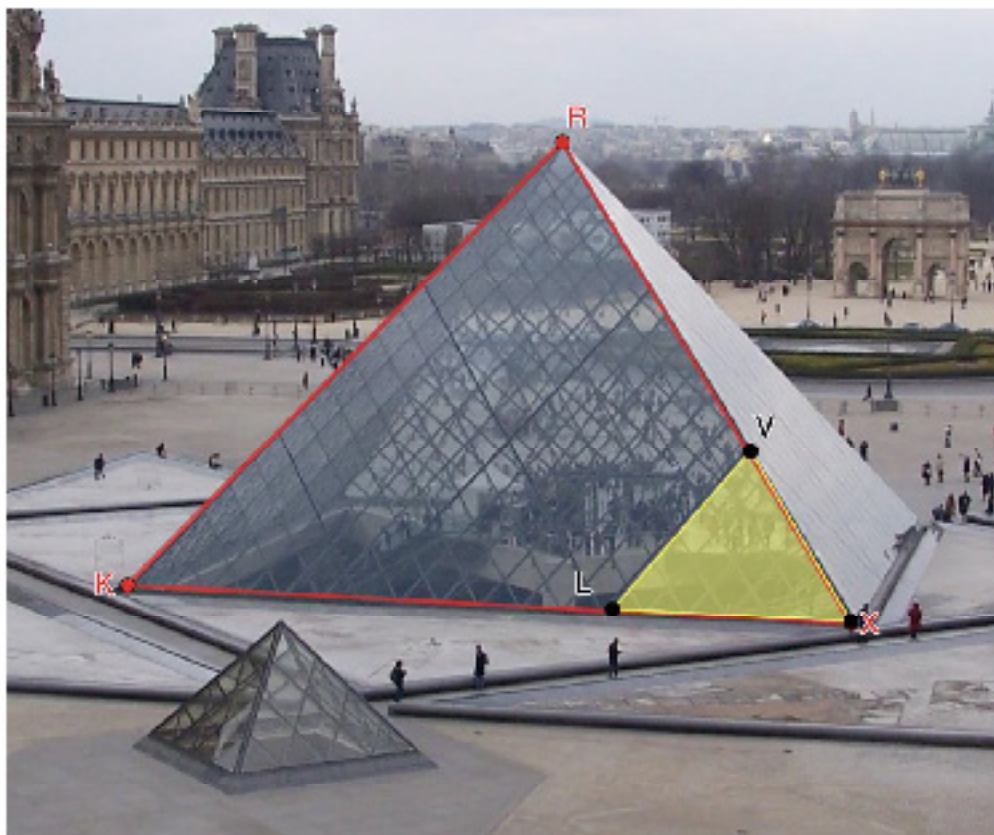
- d. How many total meters of steel were used to create one side of this bridge?

9.10.3 Collaboration: Dilations & Similar Triangles in the Real World

The Louvre in Paris is a famous museum that houses artwork like the *Mona Lisa*. Though the Louvre building was originally built in 1190 to serve as a fortress, four glass pyramids were built in the courtyard of the Louvre in 1989. These pyramids – one large one in the center, with three smaller ones along the sides – are now the defining structures of the Louvre Museum.

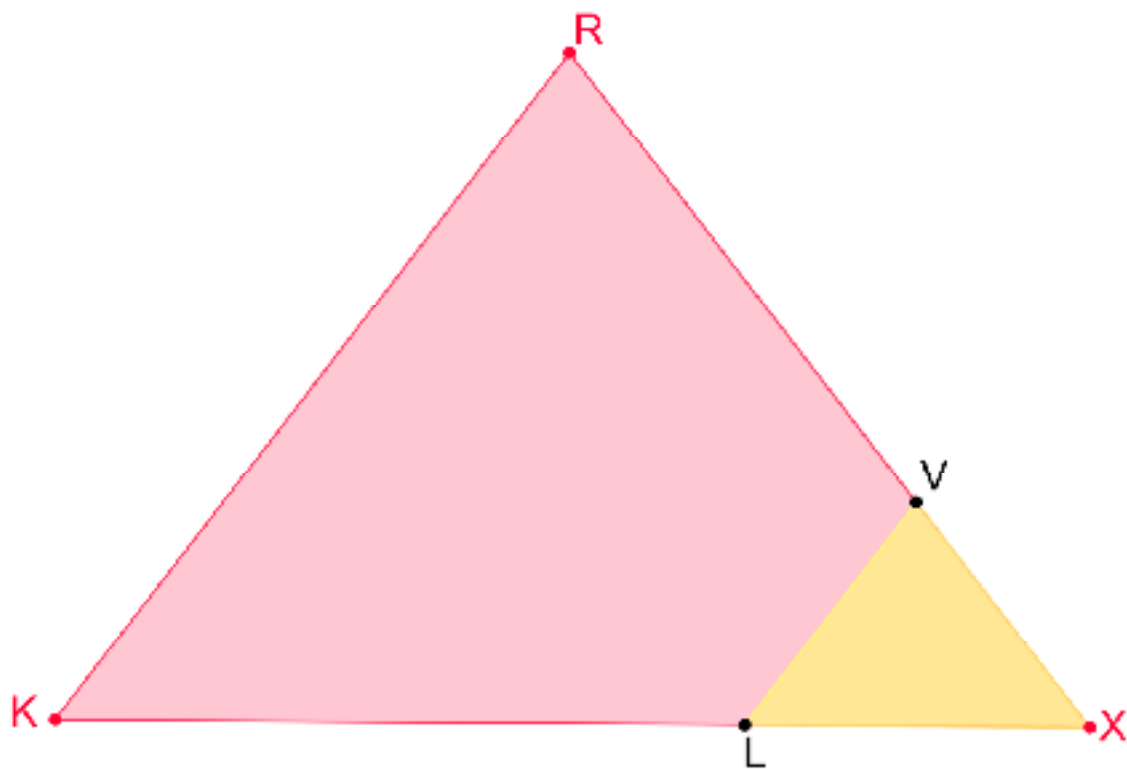


Each panel of the pyramids is built from numerous pieces of glass. The diagram shows $\triangle RKK$ and $\triangle VXL$.



Work with your partner to complete the following.

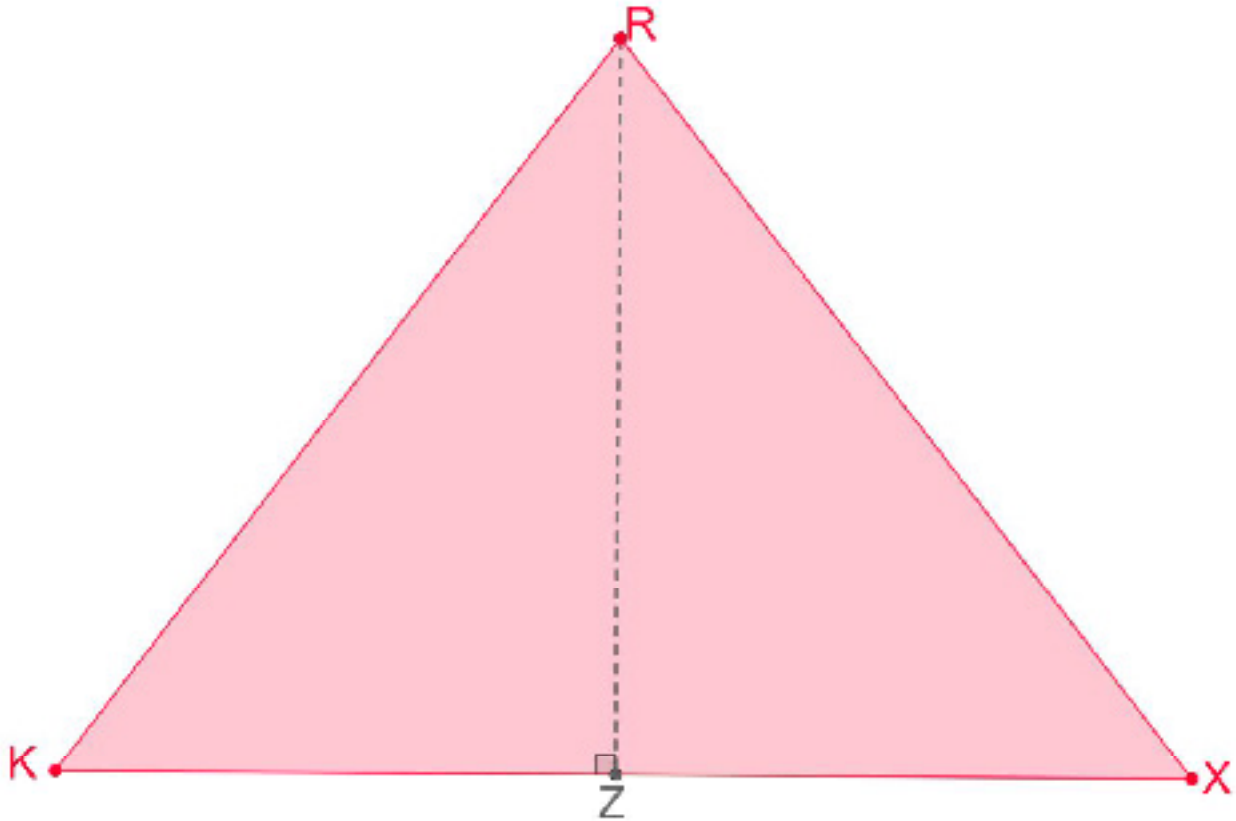
1. A scale drawing of one side panel of the large pyramid, $\triangle RXX$, is shown, including $\triangle VXL$. $RK = 106\frac{1}{2}$ feet (ft) and $LX = 37\frac{1}{2}$ ft.



- a. In this design, the full panel, $\triangle RXX$ can be determined by dilating the piece, $\triangle VXL$. Explain which point is the center of dilation.
- b. Given the scale factor of dilation is 3, find the side lengths of the full panel, $\triangle RXX$.

Side	Work	Length (in feet)
RK		
RX		
KX		

- c. Label the side lengths of $\triangle RXK$.



- d. Use the Pythagorean Theorem to find the height of the panel. Round your answer to the nearest thousandth of a foot.
- e. Find the area of the panel, in square feet. Round your answer to the nearest hundredth of a square foot.

9.10.4 Wrap-Up: Reflection

1. Complete the table.

Mindset	Choose the best description.
Did I use the partner discussions learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I put as much effort as I possibly could into the problems?	☐ Never ☐ Sometimes ☐ Usually ☐ Always

